



C23-EC-304

23132

BOARD DIPLOMA EXAMINATION, (C-23)

MARCH/APRIL—2025

DECE – THIRD SEMESTER EXAMINATION

ANALOG AND DIGITAL COMMUNICATION SYSTEMS

Time : 3 Hours]

[Total Marks : 80

PART—A

3×10=30

- Instructions :** (1) Answer **all** questions.
(2) Each question carries **three** marks.
(3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

1. Define baseband signal, carrier signal and modulated signal.
2. Define the modulation index of an AM signal.
3. Classify different types of noise.
4. Define the terms : (i) Signal to Noise Ratio. (ii) Noise Figure
5. State the need for AVC (AGC).
6. Define the terms image frequency and IMRR in a radio receiver.
7. Define PPM and its waveform.
8. List the advantages and disadvantages of digital communication system over analog communication system.
9. State the need for QAM.
10. List different types of MODEMs.

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[Contd...

PART—B

10×5=50

Instructions : (1) Answer *any five* questions.

(2) Each question carries **ten** marks.

(3) Answers should be comprehensive and criterion for valuation is the content but not the length of the answer.

- 11.** Explain the basic elements of communication system with a block diagram.
- 12.** (a) Derive the relation between total power and carrier power in AM.
(b) Explain the process of quantization with waveforms.
- 13.** Derive the time domain equation for FM signal explain its bandwidth requirements.
- 14.** Draw and explain the block diagram of indirect method of FM generation (Armstrong method).
- 15.** Draw the low-level modulated and explain its working.
- 16.** Explain the concept of time division multiplexing with block diagram.
- 17.** Explain 4QAM modulator with block diagram.
- 18.** Explain method of error correction using EFC method (Hamming Code).

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